

Deep Learning Based Discrimination of Corneal Ulcer Patterns Using Fluorescein Staining Images

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ABSTRACT

In this paper, we proposed an automatic pipeline for classifying three types of corneal ulcer patterns (point like, mixed, flaky). We first made use of H-channel related information in HSV space within cornea to perform corneal ulcers' coarse segmentation. This result was then sequentially classified using two binary classification models: 1) Point-like ulcer against mixed and flaky ulcer, and 2) mixed ulcer against flaky ulcer. The proposed pipeline was evaluated on 712 corneal ulcer images, with an accuracy of 0.858 and a kappa score of 0.796 having been obtained. We also investigated the performance of our proposed pipeline with different baseline network architectures, and identified Resnet101 to be the best-performing one.

CCS CONCEPTS

• **Computing methodologies** → **Classification and regression trees; Neural networks.**

KEYWORDS

Corneal Ulcers, Neural Networks, Image Classification

ACM Reference Format:

Zhonghua Wang, Yijin Huang, Junyan Lyu, Pujin Cheng, and Xiaoying Tang. 2020. Deep Learning Based Discrimination of Corneal Ulcer Patterns Using Fluorescein Staining Images. In *The Fourth International Symposium on Image Computing and Digital Medicine (ISICDM 2020)*, December 5–7, 2020, Shenyang, China. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3451421.3451449>

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ISICDM 2020, December 5–7, 2020, Shenyang, China

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ACM ISBN 978-1-4503-8968-6/20/12...\$15.00

<https://doi.org/10.1145/3451421.3451449>

1 INTRODUCTION

The ocular surface refers to the entire mucosal epithelium of the eyeball surface between gray lines of the upper and lower eyelid margins, and all external eye appendages maintaining the health protection system of the eyeball surface. It is composed of cornea, conjunctiva, tear film, and eyelids. The ocular surface acts as a refracting surface to accurately project light through the ocular medium to maintain cornea's optical clarity. It can protect eyes from microorganisms, trauma, and toxins. The health of the ocular surface is an essential prerequisite for maintaining clear vision. When traumatic, infectious, or endogenous pathogenic factors damage the ocular surface, they may lead to corneal diseases [9].

Fluorescein staining is the most widely used method for visually display the integrity of the ocular surface [7] and the patterns of corneal ulcers. The corneal ulcers are stained to be luminous green and the healthy corneal regions appear blue. Different types of ulcer patterns represent different disease severity levels and different treatment strategies. Distinguishing the specific ulcer pattern is of great significance for clinical decision-making.

Typically, corneal ulcers can be either point-like, flaky, or mixed (see Figure 1), representing different stages of corneal diseases. Traditionally, ophthalmologist mainly rely on visual examinations for determining the specific pattern of a corneal ulcer, such as the van Bijsterveld method [12]. However, this manner may be subjective, of low intra- and inter-rater reproducibility, as well as labor-intensive, especially because there is high similarity between neighboring categories (point-like versus mixed and mixed versus flaky) [8]. Therefore, an automatic discrimination pipeline with uniform standards is desired.

Recently, deep learning has shown its superiority and has been widely applied in medical image analysis task. [5]. In this paper, we propose an automatic pipeline making use of deep neural networks for classifying corneal ulcers of three different patterns. We firstly use H-channel related information in the HSV space for coarsely segmenting out the corneal ulcers. The segmentation result is then used for discriminating the three types of corneal ulcers (point-like,

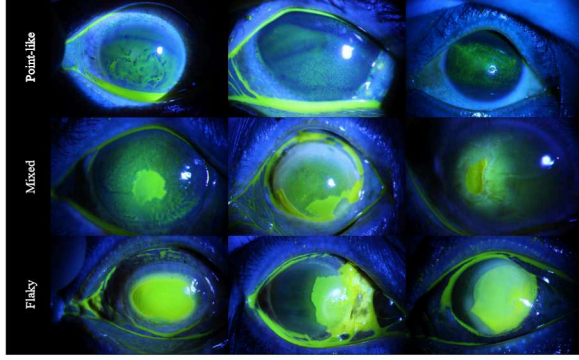


Figure 1: Representative examples of images with each of the three types of corneal ulcer patterns (point-like, mixed, flaky).

mixed, flaky) sequentially employing two binary classification models. We also test the performance of employing a single multiclass classification model to classify the three types of corneal ulcers, and compare with the proposed pipeline. Different baseline network architectures are investigated to identify the best-performing one.

2 METHOD

2.1 Coarse ulcer segmentation

Reflection areas may affect the corneal ulcer classification. We employ a previously-published method [2] to remove reflective points. Specifically, pixels with RGB intensity values satisfying $110 < R < 255$, $120 < G < 255$, $190 < B < 255$ are assigned to be of new RGB values (0, 0, 0). Then we perform coarse ulcer segmentation restricted in the cornea region. The segmentation is performed in the HSV space. In the H-channel, the intensity values of blue and green colors range between 35 and 125 (the overall H-channel ranges between 0 and 180) [1]. We keep pixels whose H-channel values lie in this range, and remove all other pixels. In this step, noisy pixels that are hard to distinguish get removed. After that, we extract pixels whose B-channel's values lie between 80 and 190 in the RGB space. These pixels are considered to be the blue background area in the segmented cornea. We remove these pixels from the segmented cornea to obtain a further segmented cornea. A further thresholding on a scaled H-channel is then applied to remove light staining residues from the segmented cornea obtained from previous steps. According to statistical analyses, the H-channel values of pixels in the segmented area lie in the range between 35 and 95. To enlarge the intensity difference between pixels, we scale the extracted H-channel intensity from range (35, 95) to range (0, 255) as follows:

$$H_{new} = (H - H_{T_{min}}) \times \frac{255}{H_{T_{max}}} \quad (1)$$

where H_{new} denotes the new value of the H-channel, $H_{T_{min}}$, $H_{T_{max}}$ respectively denote the min and max value of the original H-channel range. With the scaled H-channel, we obtain the final coarse ulcer segmentation by removing all pixels whose H-channel values lie outside the range between 5 and 250. Representative segmentation results are shown in Figure 2.

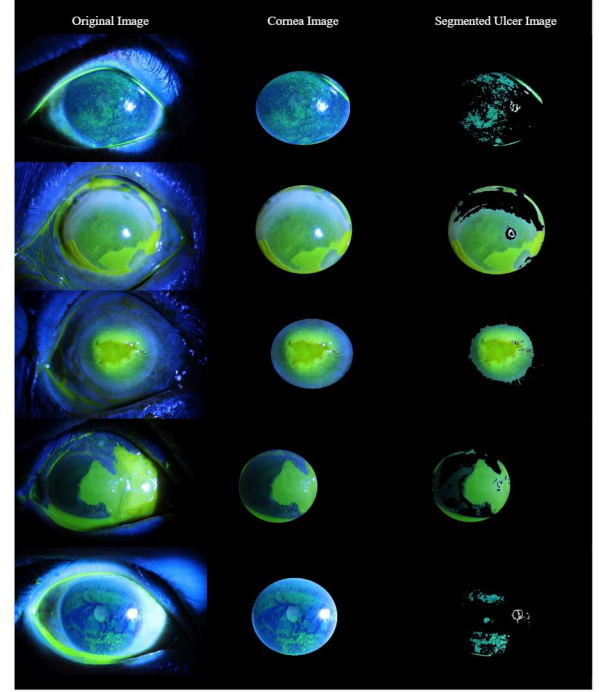


Figure 2: Representative examples of coarse ulcer segmentation.

2.2 Multi-class discrimination via binary-class discriminations

In our setting, it is challenging for a vanilla CNN model to learn discriminative features due to the high inter-class similarity between the category of mixed ulcers and its neighboring classes since it simultaneously includes point-like and flaky ulcers. Therefore, instead of using one multiclass classification model, an effective pipeline with two binary classification models is proposed to identify corneal ulcer patterns [11]. Specifically, we first employ a network to identify point-like ulcer images from all fluorescein staining images. Then another network is used to classify the rest images into mixed ulcers and flaky ulcers. Compared with a single multiclass classification model, the binary classification models perform better in their relevant tasks. In this way, the feature boundary between neighboring categories can be clarified and thus confusion can be avoided.

2.3 Overall pipeline

The overall pipeline is shown in Figure 3. The first classification model classifies point-like ulcers against mixed and flaky ulcers, and the second classification model classifies mixed ulcers against flaky ulcers. If the first classification model's result is a point-like ulcer, it will be the final result. If not, the segmented image will be sent to the second model and it will make a further classification. Please note, if the coarse ulcer segmentation result is empty, the corresponding image will be automatically classified to the category of point-like ulcers.

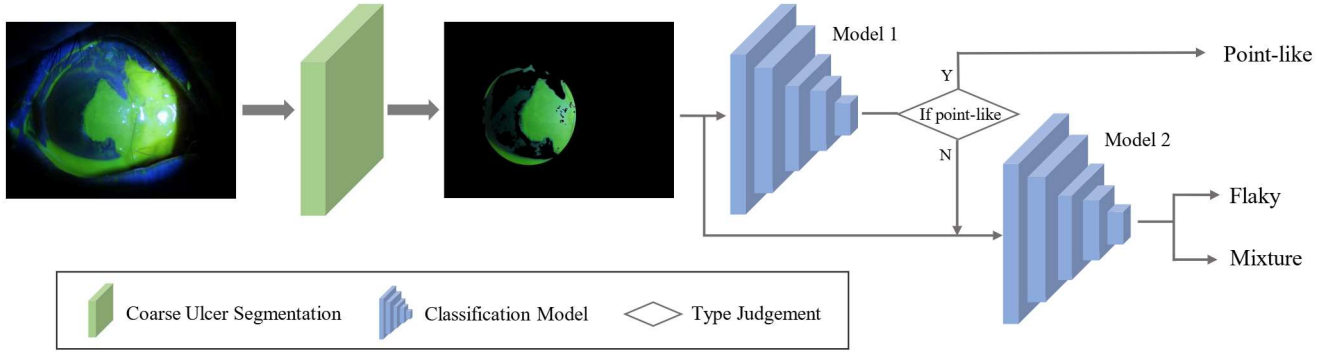


Figure 3: Flowchart of the overall pipeline.

2.4 Evaluation metrics

The most commonly used evaluation score in a classification problem is accuracy, which can intuitively reflect the proportion of correct predictions and is simple to calculate. However, each category's sample data is imbalanced in our experiments, so adjustment should be applied to address the data imbalance issue.

The kappa coefficient is a metric used for consistency testing [6]. It can also be used to measure the performance of classification. Consistency for classification problems means whether the model's predicted results are consistent with the actual classification results. With a value between -1 and 1, the kappa coefficient is obtained from the confusion matrix, as follows:

$$kappa = \frac{p_0 - p_t}{1 - p_t}, \quad (2)$$

$$p_0 = \frac{S_d}{S_t}, \quad (3)$$

$$p_t = \frac{\sum_i S_R^i \times S_C^i}{S_t^2} \quad (4)$$

where S_d denotes the sum of diagonal elements, S_t denotes the sum of all elements, S_R^i and S_C^i respectively denotes the sum of elements in the i^{th} row and the i^{th} column. The more unbalanced the confusion matrix, the lower value the kappa coefficient. The kappa coefficient avoids the situation where the accuracy is high, but the model is biased towards specific categories.

3 EXPERIMENTAL RESULTS

3.1 Dataset

The corneal ulcer dataset we used in this research comes from the SUSTech-SYSU dataset for automatically segmenting and classifying corneal ulcers [3]. The SUSTech-SYSU dataset includes 712 fluorescein staining images with corneal ulcer labels collected at Zhongshan Ophthalmic Center at Sun Yat-sen University. Further information of that dataset can be found in our previous work [3]. All images were divided into 64% training dataset, 16% validation dataset, and 20% testing dataset through five-fold cross-validation.

3.2 Experimental setup

We experimented on two types of deep CNN architectures in this work. 1) Residual Neural Network (Resnet) [4] was proposed by Kaiming He and others. Using residual modules, Resnet solves the "degeneration" problem of deep neural networks to a certain degree. It has been successfully applied in different medical image segmentation and classification tasks and has achieved promising results. 2) The Inception model proposed by Google [10] is another series of network structures used for classification problems. Using multiple branches in the Inception module, the classification network's accuracy is improved via the fusion of multiple receptive field information while the model's computational cost and memory cost is fully reduced.

All CNNs were implemented utilizing Pytorch, and all training and testing experiments were conducted on a workstation equipped with RTX 2080Ti GPU. For each model, we trained for a total of 30 epochs with a batch size of 16, and the model was initialized with parameters pretrained on ImageNet. An ADAM optimizer with a learning rate of 0.0001 and a cosine decay was adopted. The input image size was 256×256 .

3.3 Results

Figure 4 compares the performance between the single multiclass classification model and our proposed pipeline with different baseline network architectures. As seen in Figure 4, the proposed pipeline achieved better results with all different baseline network architectures. With Resnet101-based network, the proposed pipeline had a 2.1% improvement over the multiclass classification. Such a result indicates that more detailed information is learned by two sequential classification models.

For further evaluations of the proposed pipeline, we analyzed the impact of the input image by comparing the classification performance of the proposed pipeline with two different types of input images, namely the cornea image and the segmented ulcer image, the comparison result of which are shown in Figure 5. As seen in Figure 5, using H-channel based coarse ulcer segmentation achieved better classification results with both Resnet50 and Resnet101. However, the results with cornea images being the input were better using Inception-v3. These results indicate that the effectiveness of

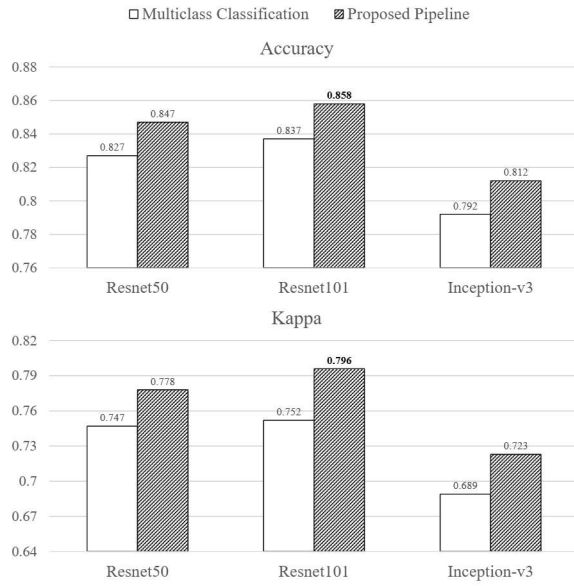


Figure 4: Comparison of the proposed pipeline with the single multiclass classification model using segmented ulcer images with different baseline network architectures.

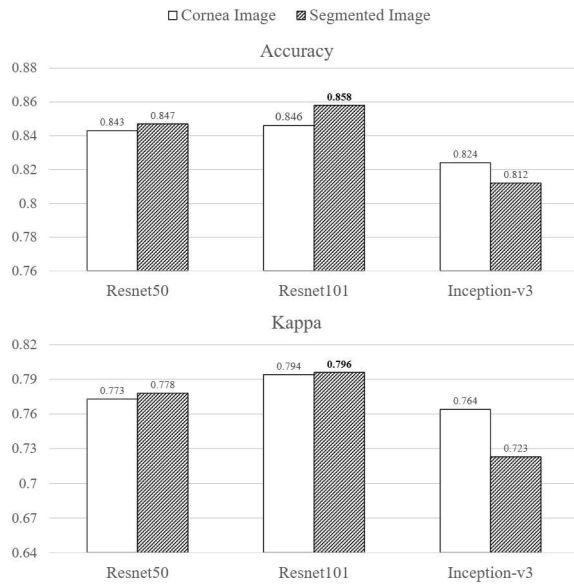


Figure 5: Comparison of the proposed pipeline using cornea images and segmented ulcer images with different baseline network architectures.

the coarse ulcer segmentation is related to the baseline network architecture selected. With a higher generalization ability, Resnet can learn more complex information from the segmented ulcer images to enhance the classification results. Overall, the proposed pipeline using coarsely segmented ulcer images, with Resnet101 being the network architecture, obtained the best result.

4 CONCLUSION

This paper proposed a pipeline making use of deep learning to classify three types of corneal ulcer patterns. A thresholding method in the HSV color space was adopted to generate coarse segmentation of the ulcer area. Instead of one single multiclass model, two binary classification models were designed to sequentially identify the three types of corneal ulcer patterns. The effectiveness of the proposed pipeline has been successfully validated on the publicly-available SUSTech-SYSU dataset.

With that being said, this work still has limitations. The threshold for segmenting the ulcer within cornea was determined empirically, which may not be optimal for all images. Moreover, simple linear scaling was applied in the H-channel in the coarse ulcer segmentation step, which may induce a loss of ulcer information.

An adaptive thresholding method is one of our future works. Furthermore, investigating new methods for training on imbalanced and small-sampled datasets will also be our future research plan.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (62071210), the Shenzhen Basic Research Program (JCYJ20190809120205578), the National Key RD Program of China (2017YFC0112404), and the High-level University Fund (G02236002). The authors declare that they have no competing financial interests.

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